Campus Log Anomaly Detection under Proxy Labels: Validity Limits of Archive Splits and Masked TCN–Transformers

Hoang Tu Trinh[1⋆], Tien Thanh Cao[1], and Manh Ha Tran[1]

Faculty of Information Technology,
Ho Chi Minh City University of Foreign Languages – Information Technology,
Ho Chi Minh City, Vietnam
23dh113972@st.huflit.edu.vn, thanhct@huflit.edu.vn, hatm@huflit.edu.vn

Abstract. Archive packaging and heuristic proxy labels can make log anomaly evaluation look stronger than ranking warrants. On selective HUFLIT campus backup extracts we treat evaluation validity as the primary object of study, comparing (A) an archive-order stream split with validation contamination against (B) a jointly decontaminated protocol (exact-line deduplication, source-bounded windows, backup-day holdout) under HTTP/keyword *proxy* labels—not SOC ground truth. Three findings dominate: OR window labels inflate prevalence (Protocol A test rate 80.5%; always-anomaly F1≈0.892); switching protocols changes detector rankings and lifts; and a surveyed masked TCN–Transformer adds leave-one-out scoring latency without a reliable accuracy gain on these extracts. We report AUPRC lift over prevalence within each protocol, trivial baselines, and label-aggregation sensitivity; we do *not* claim operational incident detection.

Keywords: AIOps · Log Anomaly Detection · Proxy Labels · Experimental Validity · TCN–Transformer · Campus Infrastructure.

1 Introduction

Campus IT services emit heterogeneous operational logs. Sequence detectors can score unusual template patterns [3–6], but archive packaging, duplicate backups, and heuristic labels can invent "temporal" structure that models then appear to learn. Prior empirical studies already show that training-data selection, grouping, class distribution, and noise strongly affect public LAD benchmarks [10, 11]. We extend that validity agenda to a setting those studies do not isolate: institutional *backup archives* whose folder order is not event time, with exact raw-line contamination audits, OR-window prevalence inflation, and proxy-label/feature coupling.

This paper treats those threats as first-class experimental objects. On HUFLIT selective extracts we ask three research questions: **(RQ1)** How much

⋆ Corresponding author: 23dh113972@st.huflit.edu.vn

do OR proxy window labels inflate prevalence and trivial F1 relative to stricter k-of-L aggregations? **(RQ2)** How do detector conclusions change between an archive-order split (Protocol A) and a jointly decontaminated backup-day protocol (Protocol B), and what does removing only exact-line deduplication change? **(RQ3)** Under these proxy labels, does adding a causal TCN to a masked Transformer improve ranking enough to justify its leave-one-out scoring cost? We survey a masked TCN–Transformer and a no-TCN ablation as *evaluation targets*, not as a claimed architecture advance. We add trivial baselines (always-anomaly; no-skill AUPRC = prevalence), report AUPRC lift = AUPRC − test prevalence *within* each protocol, and treat validation-selected F1 thresholds as *diagnostic* operating points only.

The contributions are:

- **Validity protocol on campus archives**: measure OR-window prevalence inflation against an always-anomaly trivial baseline, audit exact raw-line and template-window contamination, and retain reproducible dual-protocol evaluation artifacts under heuristic proxy labels (not SOC ground truth).
- **Joint decontamination protocol**: Protocol B applies exact-line SHA-1 dedup, windows that never cross (backup-day, source, member), and backup-day holdout; we add a single-factor note on B without dedup and label-aggregation sensitivity (k-of-L). We do *not* claim a full causal factorial over all three factors.
- **Threshold-aware reading**: fixed $\mu+3\sigma$ versus validation-selected F1 thresholds with precision/recall, separating ranking from calibration; validation-tuned F1 is not a deployable operating point.
- **Surveyed TCN evidence**: on these extracts and settings the TCN does not consistently beat a masked Transformer without TCN, while increasing leave-one-out scoring latency; keyword/status ablation shows only partial proxy-token leakage into PCA lift.

Relation to an accepted companion study. A related manuscript on network-aware event-sequence modeling with Drain3 parsing and GraphWalk transition scoring has been accepted at CSoNet 2026 [1]. That work targets cross-dataset detector comparison (HDFS, BGL, HUFLIT-Career) under a graph-walk formulation. The present paper addresses a different question on institutional *backup archives*: how archive ordering, exact-line contamination, OR-window proxy labels, and jointly decontaminated day holdout change validity conclusions for masked TCN–Transformers. We use a custom similarity parser (not Drain3), do not claim GraphWalk contributions, and do not reuse CSoNet tables or primary claims.

2 Related Work

2.1 Log Parsing and Template Extraction

Log parsing transforms free-form messages into event templates by separating constants from dynamic parameters [2, 13]. Drain uses a fixed-depth parse tree [2];

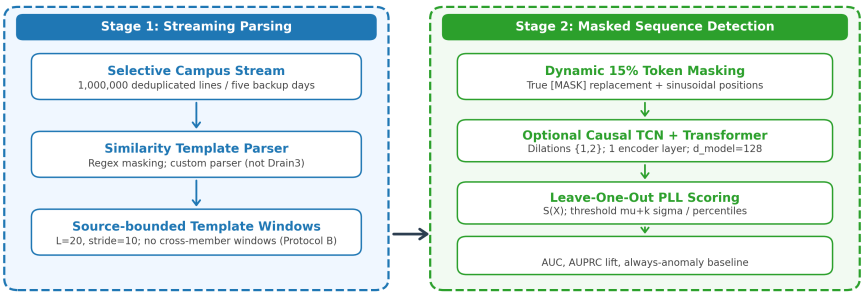


Fig. 1. Surveyed detector pipeline: streaming parsing and masked pseudo-log-likelihood (PLL) scoring.

Spell uses streaming sequence matching [13]. Our artifact does *not* implement Drain or Drain3. It applies regex masking and a linear scan over equal-length templates, merging tokens when positional similarity exceeds 0.5. We call this the *custom similarity parser* and avoid attributing Drain3 properties or throughput to it.

2.2 Deep Learning for Log Anomaly Detection

Sequence-based anomaly detection models learn temporal transition dynamics over sequences of log templates [3, 4, 7]. Du et al. [3] introduced DeepLog (LSTM next-token prediction). Meng et al. [4] proposed LogAnomaly with semantic template vectors. SwissLog [6] and LogRobust [5] address instability and interleaved threads. Classical reconstruction baselines such as PCA on count vectors remain competitive on some structured logs [7]. Recent Transformer–TCN hybrids [16], multi-scale semantic models [14], and contrastive masked objectives [15] continue this line. Public corpora such as Loghub [12] remain the standard sanity setting; we defer a full HDFS/BGL re-run under the present harness to future work.

2.3 Experimental Validity in Log Anomaly Detection

Le and Zhang [10] show that training-data selection, grouping, class distribution, data noise, and early-detection settings can overturn optimistic LAD conclusions on public benchmarks. Albert [11] argues that training should precede testing chronologically and analyzes grouping, anomaly ratio, and duplicate sequences. Our contribution is complementary rather than redundant: we study an institutional campus *backup archive* where packaging order is not wall-clock event time; we report exact raw-line hash contamination and OR-window prevalence inflation with an always-anomaly baseline; we probe proxy-label/feature coupling via template ablation and k-of-L sensitivity; and we survey a masked TCN–Transformer under that archive protocol without claiming a general architectural verdict.

Table 1. Qualitative comparison of related detectors. Latency is *not* cross-comparable: this work reports batch-1 leave-one-out PLL; DeepLog/LogAnomaly-lite legacy batched scoring is NR for batch-1.

Approach	Parser	Detector	Focus	CPU ms
DeepLog [3]	Spell	LSTM next-token	Sequence LAD	NR[†]
LogAnomaly [4]	FT-Tree + template2Vec	LSTM (seq. + quant.)	Sequence LAD	NR[†,‡]
LogRobust [5]	Drain	Bi-LSTM+Attn	Unstable logs	NR
SwissLog [6]	Online	Attentive Bi-LSTM	Interleaved logs	NR
Le & Zhang [10]	Drain	Multiple DL	Public-benchmark validity	NR
Albert [11]	Drain	DeepLog-based	Chronological LAD protocol	NR
This work	Custom similarity	Masked TCN–Transformer	Campus-archive validity	51.1[§]

[†]Legacy batched timing is retained in evaluation artifacts but is not relabeled as batch-1 PLL. [‡]LogAnomaly-lite is our reimplementation on indexed templates and does not reproduce the original template2Vec semantic representation or the quantitative (count) branch; scoring uses mean token reconstruction NLL only. [§]Batch-1 leave-one-out PLL on Protocol B (CPU, 2 threads).

3 Methodology

3.1 System Architecture Overview

The measured system has two stages: (i) **Parsing**: selected syslog messages are tokenized by the custom similarity parser; (ii) **Masked sequence scoring**: sliding template windows are corrupted by true token replacement, encoded by an optional causal TCN plus Transformer, and scored by leave-one-position-out *pseudo-log-likelihood* (PLL): the mean negative log probability of each non-padding token when that position alone is masked.

3.2 Masked TCN-Transformer Denoising Sequence Model

We survey a masked template language model as an evaluation target under proxy labels (not a claimed SOTA detector): 15% of input tokens are replaced by a reserved mask token, representations pass through a Temporal Convolutional Network (TCN) [8] and Transformer self-attention [9], and a linear vocabulary head predicts the original identities only at selected positions.

Embedding and Dynamic Masking A sliding window of length $L = 20$ with stride $S = 10$ extracts template sequences $\mathbf{X} = [T_1, T_2, \ldots, T_L]$. During training, 15% of tokens are replaced with `[MASK]` ($ID = |\mathcal{V}| + 1$), distinct from padding ($ID = 0$). Padding masks are applied in self-attention. Embeddings use sinusoidal positional encodings [9]:

$$\mathbf{E} = \text{Embedding}(\mathbf{X}) + \mathbf{P}_{\text{pos}} \in \mathbb{R}^{L \times d_{\text{model}}} \tag{1}$$

Temporal Convolutional Network (TCN) Layer Dilated causal convolutions [8] with $d \in \{1, 2\}$ extract local patterns. The TCN branch

uses causal convolutions; the subsequent Transformer encoder remains bidirectional, as intended by the masked-PLL objective:

$$\mathbf{y}(t) = (\mathbf{x} *_d f)(t) = \sum_{i=0}^{k-1} f(i) \cdot \mathbf{x}(t - d \cdot i) \tag{2}$$

$$\mathbf{H}_{\text{TCN}} = \text{LayerNorm}(\text{Proj}(\mathbf{E}) + \text{GELU}(\text{CausalTCN}(\mathbf{E}))) \tag{3}$$

Transformer Encoder and Scoring A single Transformer encoder layer (pre-LN, multi-head attention, GELU FFN) contextualizes the sequence. Training minimizes masked cross-entropy. Let $\mathcal{V}$ be the set of non-padding positions and

$$\mathbf{X}^{(-t)} = [T_1, \ldots, \texttt{[MASK]}_t, \ldots, T_L]. \tag{4}$$

Inference uses leave-one-position-out PLL:

$$S(\mathbf{X}) = \frac{1}{|\mathcal{V}|} \sum_{t \in \mathcal{V}} -\log P(T_t \mid \mathbf{X}^{(-t)}). \tag{5}$$

The primary threshold is $\tau = \mu_{\text{val}} + 3\sigma_{\text{val}}$ on nominal validation scores; as a diagnostic we also report a validation-selected F1 threshold applied to test scores. That diagnostic threshold is fixed before looking at test labels and is *not* presented as a validated deployment operating point.

4 Experimental Evaluation

4.1 Corpus and Window Labels

We stream selective June daily backup archives from a local institutional campus extract, focusing on application/service syslog members from aca, careerhub, and courses. We skip FortiGate CSV members (heterogeneous firewall schema; not comparable to the syslog template stream without a separate parser) and members >500 MiB (computational bound for the selective ingest). Protocol B keeps the first 1M unique raw lines under per-day and per-file caps (200k lines/day; 40k/file). Proxy line labels use HTTP 4xx/5xx, deny/block fields, and keywords—not SOC adjudication. Unless noted, a window is positive iff *any* of its L lines is positive (OR aggregation). Under independence, $1 - (1 - p)^L$ with p=0.0796 predicts prevalence ≈ 0.81, matching Protocol A's 80.5% test rate. Applying the same raw-line rule that *defines* proxy labels as if it were a detector yields perfect agreement by construction; we therefore treat that check only as a proxy-label consistency probe, not as an independent detector.

4.2 Protocol A (archive-order split with validation contamination)

Records are ordered by backup day → ZIP → member → in-member position, then split 80/10/10 before windowing. This is *not* event time: windows may cross

file/service boundaries. A raw-line hash audit found 36.235% of unique validation lines also in training and 11.382% of unique test lines also occur in validation; train/test exact raw-line overlap is only 0.0176% of unique test lines (13 shared hashes), so Protocol A's primary risk is archive-order structure and *validation contamination* (validation is used for threshold calibration) rather than massive train→test line leakage. Exact template-window overlap is likewise small on test (13 of 7,573 unique test windows).[1]

4.3 Protocol B (dedup + source-bounded + backup-day holdout)

Protocol B *jointly* applies exact-line deduplication, boundary-safe windowing, and backup-day holdout; we therefore treat A→B as a comparison of protocols, not a causal ablation of archive ordering alone. We keep the first SHA-1 of each *raw line* (262,194 duplicates skipped while filling 1M unique lines), emit windows only inside one (backup-day, source, member) group, and assign backup days 15–17 Jun→train, 18 Jun→val, 19 Jun→test. Day folders are backup collection periods, not verified per-event timestamps. Test-window prevalence is 56.36% (always-anomaly F1 baseline 0.721).

Global exact-line deduplication may also remove legitimate repeated events that recur identically across backup days; we therefore treat it as a *conservative decontamination* step that can alter the backup-day-held-out test distribution, and report this as an additional validity limitation rather than a claim that every duplicate is a backup artifact. Holding day split and source-bounded windows fixed, skipping exact-line deduplication retains those 262,194 duplicate raw lines in the stream fill toward the 1M cap and therefore changes which later unique lines enter the day-held-out partitions; we report this as a single-factor distributional caveat rather than a full re-leaderboard (full factorial ablations remain future work).

Raw-line dedup does *not* imply unique template windows. Exact template-ID window sequence overlap (SHA-1 of the integer window vector): 505 of 11,065 distinct validation windows (4.56%) also occur in training, and 110 of 11,137 distinct test windows (0.99%) also occur in training.[2]

4.4 Experimental Setup

Table 2 lists corpus settings for Protocol B. Table 3 summarizes shared baseline configurations. Protocol A uses the same architectures on the earlier stream-position arrays. Training is nominal-only (proxy labels select the nominal subset and the nominal validation scores used for $\mu+3\sigma$); seeds $\{42, 101, 202, 303, 404\}$; hardware is CPU (PyTorch 2.8.0+cpu, 2 intra-op threads). Masked models use AdamW (10^{-3}, weight decay 10^{-4}), batch size 256, 8 epochs,

[1] Derived evaluation artifacts and split metadata are retained in the project repository for co-author reproduction.

[2] Split-overlap statistics appear in the dual-protocol evaluation metadata retained with the code artifact.

gradient clip 1.0, and early stop on nominal validation PLL when available; the encoder is one pre-LN Transformer layer (d=128, 4 heads, FFN 256) unless noted as an ablation target.

Table 2. Protocol B corpus setup. Window label = OR over L proxy line labels.

Item	**Value**
Unique lines kept / dups skipped	1,000,000 / 262,194
Train / val / test windows	59,966 / 19,988 / 19,990
Test positives / prevalence	11,267 / 56.36%
Always-anomaly F1 baseline (test)	0.721
Vocab (templates+pad+mask)	1,183
L, stride	20, 10
Seeds	$42, 101, 202, 303, 404$
Hardware	CPU, 2 PyTorch threads

4.5 Models and Metrics

We evaluate five detector families: PCA count-vector reconstruction [7], a DeepLog-style LSTM [3], LogAnomaly-lite [4], a masked Transformer without TCN, and a masked TCN–Transformer. Both masked models are evaluated under the two protocols on the same window arrays within each protocol. Metrics: F1/precision/recall at $\mu+3\sigma$, ROC-AUC, AUPRC, and **AUPRC lift** = AUPRC − test prevalence (interpreted *within* each protocol, not as a cross-prevalence normalized score), always-anomaly F1 baseline, and batch-1 PLL latency (masked models only).

Table 3. Baseline and masked-model configuration (both protocols unless noted).

Component	Configuration
PCA	Count vectors; n=32 components (capped); PCA (`sklearn`; SVD solver); score = reconstruction MSE.
DeepLog-style*	Emb 64, LSTM hidden 128, 2 layers; next-token CE; score = mean next-token NLL; Adam 10^{-3}; batch 256; 6 epochs.
LogAnomaly-lite	Emb+2-layer LSTM recon on template IDs only (no template2Vec; no quantitative branch); score = mean token NLL; same optimizer.
Masked Transformer	d=128, heads 4, 1 layer, FFN 256, mask 15%; AdamW 10^{-3}; batch 256; 8 epochs; score = leave-one-out PLL.
Masked TCN–Transformer	As above + causal conv k=3, dilations $\{1, 2\}$; same optimizer and PLL score.
Primary / diagnostic thr.	$\mu_{\text{val}}+3\sigma_{\text{val}}$ on nominal val; validation-F1-selected threshold, fixed before test evaluation (diagnostic only).

*Our DeepLog-style reimplementation uses mean next-token NLL for continuous ranking and threshold evaluation rather than the original top-g decision rule [3]. Lift = AUPRC − test prevalence within the same protocol.

4.6 Results

On Protocol A, the DeepLog-style LSTM leads ROC-AUC (0.781); LogAnomaly-lite is close (0.777). Under these extracts and settings, the surveyed TCN is *worse* than no-TCN on mean F1/AUC and much slower at batch-1 PLL. The always-anomaly F1 baseline (0.892) exceeds every learned F1 at $\mu+3\sigma$; AUPRC must be read against prevalence 0.805 (lift = AUPRC − prevalence within Protocol A).

On Protocol B (Table 5), DeepLog-style and LogAnomaly-lite retain the strongest mean AUPRC lifts (+0.138 and +0.112), but with large seed variance (DeepLog-style ROC-AUC 0.657±0.134). Masked models lose backup-day ranking (AUC≈ 0.49–0.53). PCA keeps a smaller but more stable lift (+0.098). Table 8: $\mu+3\sigma$ starves recall for most detectors (PCA/DeepLog-style/masked Transformer recall ≈ 0–0.01), while thresholds selected on validation and evaluated on *test*

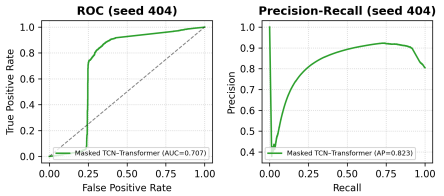


Fig. 2. Protocol A ROC/PR curves for the Masked TCN–Transformer, seed 404 (illustrative; one of five evaluation seeds; tables report mean±std).

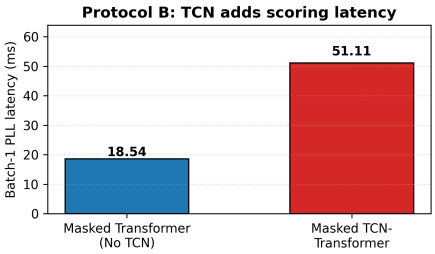


Fig. 3. Protocol B batch-1 PLL latency: Masked Transformer (No TCN) vs. Masked TCN–Transformer.

recover F1≈ 0.73–0.75—near the always-anomaly test baseline (0.721)—showing that much of the apparent F1 gain is driven by prevalence and threshold calibration rather than strong ranking quality. Under Protocol B, the surveyed TCN remains slower (51 vs. 19 ms) without consistent accuracy gains on this held-out backup day.

Table 6 summarizes label-aggregation sensitivity. Replacing OR stored labels with a 1-of-L trigger-template count nearly halves Protocol B test prevalence (0.564 $\rightarrow$ 0.276) and always-anomaly F1 (0.721 $\rightarrow$ 0.433), confirming that aggregation choice is a first-order driver of trivial baselines. Raising k to 2 or 3 barely changes prevalence further because trigger-positive windows usually contain multiple matching positions. A rule that reapplies the same k-of-L definition used to build those labels achieves F1= 1 by construction; we report it only as a proxy-consistency check.

Table 7: remapping 167 proxy-trigger templates ($\approx$ 9.3% of test tokens) to padding in train/val/test, then refitting PCA and retraining the masked Transformer, reduces PCA lift from +0.098 to +0.084 without collapsing ranking to chance, while the masked Transformer stays near chance before and after ablation. This is therefore a *retrained* sensitivity analysis rather than a scoring-time-only probe. Trigger templates are those whose template string matches a status/keyword pattern used by the proxy-labeling rules (HTTP 4xx/5xx and deny/block/error-like tokens). For the masked models, remapping to padding removes these positions from attention and from the PLL average rather than introducing a dedicated ablation token; for PCA, pad IDs are simply omitted from the count vector. Residual correlation remains after template-level trigger ablation; without SOC labels, it cannot be interpreted as validated operational anomaly signal.

Due to privacy restrictions, raw campus telemetry cannot be redistributed. We retain code, configuration, aggregate split statistics, and derived evaluation artifacts in the project repository for co-author reproduction; a public mirror is contingent on institutional approval.

Table 4. Protocol A (archive-order stream-position with validation contamination): $\tau = \mu{+}3\sigma$; test prevalence 80.5%; always-anomaly F1 baseline$\approx$ 0.892; AUPRC no-skill$\approx$ 0.805

Model	F1	ROC-AUC	AUPRC	Lift	Batch-1 ms
Always-anomaly (trivial)	0.892	0.500	0.805	0.000	—
PCA	0.000 ± 0.000	0.738 ± 0.004	0.913 ± 0.003	+0.108	NR
DeepLog-style	0.011 ± 0.022	$\mathbf{0.781 \pm 0.020}$	0.882 ± 0.025	+0.077	NR
LogAnomaly-lite	0.010 ± 0.013	0.777 ± 0.045	0.882 ± 0.040	+0.077	NR
Masked Transformer (No TCN)	0.348 ± 0.410	0.769 ± 0.035	0.870 ± 0.036	+0.065	2.04
Masked TCN–Transformer	0.040 ± 0.016	0.723 ± 0.018	0.836 ± 0.014	+0.031	52.92

Table 5. Protocol B (dedup + source-bounded + backup-day holdout): primary $\tau = \mu{+}3\sigma$; test prevalence 56.4%; always-anomaly F1 baseline 0.721

Model	F1@3σ	ROC-AUC	AUPRC	Lift	Batch-1 ms
Always-anomaly (trivial)	0.721	0.500	0.564	0.000	—
PCA	0.000 ± 0.000	0.629 ± 0.008	0.661 ± 0.006	+0.098	—
DeepLog-style	0.013 ± 0.027	0.657 ± 0.134	0.702 ± 0.153	$\mathbf{+0.138}$	NR
LogAnomaly-lite	0.385 ± 0.293	0.632 ± 0.109	0.676 ± 0.131	+0.112	NR
Masked Transformer (No TCN)	0.001 ± 0.002	0.493 ± 0.009	0.511 ± 0.007	−0.052	18.54 ± 0.47
Masked TCN–Transformer	0.151 ± 0.183	0.532 ± 0.112	0.557 ± 0.124	−0.007	51.11 ± 9.08

Table 6. Protocol B label-aggregation sensitivity on the held-out test day (L=20). OR uses stored proxy window labels. k-of-L counts templates matching the same keyword/status patterns as the proxy rules (167 triggers). Rule-oracle F1= 1 reapplies the labeling definition and is a consistency check only. PCA lift = AUPRC − prevalence under that label definition (seed 42).

Aggregation	Prev.	Always-anom. F1	PCA AUC	PCA lift	Rule-oracle F1
OR (stored proxy)	0.564	0.721	0.619	+0.090	—
1-of-L (trigger)	0.276	0.433	0.825	+0.302	1.000
2-of-L (trigger)	0.276	0.432	0.825	+0.303	1.000
3-of-L (trigger)	0.274	0.430	0.825	+0.303	1.000

Table 7. Protocol B keyword/status template ablation: 167 trigger templates ($\approx$ 9.3% of test tokens) are remapped to padding in train/val/test; PCA is refit and the masked Transformer is retrained on the ablated arrays before scoring

Model	ROC-AUC	AUPRC lift	Δ lift vs. full
PCA (full)	0.629 ± 0.008	+0.098	—
PCA (ablated)	0.626 ± 0.007	+0.084	−0.014
Masked Transformer (full)	0.493 ± 0.009	−0.052	—
Masked Transformer (ablated)	0.494 ± 0.005	−0.050	+0.002

Table 8. Protocol B threshold comparison (mean over 5 seeds): fixed $\mu+3\sigma$ vs. *test* metrics at a validation-selected F1 threshold τ_{val} (diagnostic)

Model	F1@3σ	P@3σ	R@3σ	F1@τ_{val}	P@τ_{val}	R@τ_{val}
PCA	0.000	0.000	0.000	0.729	0.585	0.966
DeepLog-style	0.013	0.423	0.007	0.745	0.619	0.934
LogAnomaly-lite	0.385	0.625	0.297	0.742	0.601	0.969
Masked Transformer	0.001	0.152	0.000	0.745	0.620	0.934
Masked TCN–Transformer	0.151	0.354	0.102	0.741	0.608	0.949

5 Discussion & Threats to Validity

What sequence means: Protocol A orders archive members, not wall-clock events. Protocol B jointly removes cross-member windows and duplicate raw lines but still uses backup-day folders rather than verified timestamps. Seed variance does not capture day-to-day sampling uncertainty because Protocol B contains a single held-out backup day; we did not claim multi-day generalization.

Window OR labels: OR aggregation manufactures high prevalence and makes the always-anomaly F1 baseline a necessary reference (not an upper bound: a perfect detector still has F1= 1). Ranking metrics and lift-over-prevalence must accompany thresholded F1; AUPRC lift = AUPRC − test prevalence is interpreted within each protocol and not directly compared across prevalences.

Proxy labels and label–feature coupling: HTTP status tokens, deny/block fields, and keyword heuristics both *define* proxy labels and often remain visible in template text. Proxy labels also select nominal-only training windows and the nominal validation scores used for $\mu+3\sigma$. Models may therefore learn the labeling rule rather than operational anomalies. Our keyword/status template ablation (Table 7) and k-of-L sensitivity (Table 6) support prevalence/leakage analysis without claiming residual operational signal. Gold SOC labels remain required before operational claims.

Parser: The custom parser has no gold template annotations; the 0.5 merge threshold lacks a full sensitivity sweep in this revision, and parser merge errors may alter event frequency and sequence structure [10].

Corpus selection: Skipping FortiGate CSV and >500 MiB members is a deliberate selective-ingest policy (schema mismatch and compute bound), not a claim of full-archive coverage.

TCN scope: Negative evidence for the surveyed one-layer masked TCN–Transformer applies to these HUFLIT extracts, proxy labels, and training settings; it is not a general claim that TCN architectures cannot help elsewhere.

Ethics: Campus telemetry stays private; any future release needs redaction and authorization.

6 Conclusion

We framed HUFLIT campus-log evaluation as a *validity* study under heuristic proxy labels. OR window labels inflate prevalence and make thresholded F1 look

stronger than ranking warrants; a jointly decontaminated backup-day protocol changes conclusions relative to an archive-ordered stream split; and the surveyed masked TCN–Transformer adds leave-one-out latency without a reliable benefit on the evaluated Protocol B held-out day. Label-aggregation sensitivity shows that moving from OR to $k \geq 2$ sharply reduces trivial baselines. We do not claim SOC-verified incident detection or multi-day generalization. Future work: fuller factorized protocol ablations, additional backup-day folds, event timestamps, SOC labels, parser-threshold sweeps, and Loghub [12] sanity checks under the same harness.

Acknowledgment

The authors thank the Faculty of Information Technology, Ho Chi Minh City University of Foreign Languages - Information Technology (HUFLIT), Ho Chi Minh City, Vietnam for providing the production campus log telemetry and computing infrastructure. Generative AI tools were used for literature discovery, consistency checking, and language refinement. All cited sources were independently verified by the authors. Study design, implementation, experimental execution, statistical analysis, interpretation, and final scholarly judgments were performed and approved by the authors.

Disclosure of Interests

The authors have no competing interests to declare that are relevant to the content of this article.

References

1. Cao, T.T., Trinh, T.H., Tran, H.M.: Network-Aware Event Sequence Modeling for User-Behavior and System Log Anomaly Detection. In: Proc. 15th Int. Conf. Computational Science and Network (CSoNet 2026), accepted for publication (2026).

2. He, P., Zhu, J., Zheng, Z., Lyu, M.R.: Drain: An Online Log Parsing Approach with Fixed Depth Tree. In: Proc. IEEE Int. Conf. Web Serv. (ICWS), pp. 33–40 (2017).

3. Du, M., Li, F., Zheng, G., Srikumar, V.: DeepLog: Anomaly Detection and Diagnosis from System Logs through Deep Learning. In: Proc. ACM SIGSAC Conf. Comput. Commun. Secur. (CCS), pp. 1285–1298 (2017).

4. Meng, W., Liu, Y., Zhu, Y., et al.: LogAnomaly: Unsupervised Detection of Sequential and Quantitative Anomalies in Unstructured Logs. In: Proc. 28th Int. Joint Conf. Artif. Intell. (IJCAI), pp. 4739–4745 (2019). https://doi.org/10.24963/ijcai.2019/658

5. Zhang, X., Xu, Y., Lin, Q., et al.: Robust Log-Based Anomaly Detection on Unstable Log Data. In: Proc. 27th ACM Joint Eur. Softw. Eng. Conf. Symp. Found. Softw. Eng. (ESEC/FSE), pp. 807–817 (2019).

6. Li, X., Chen, P., Jing, L., He, Z., Yu, G.: SwissLog: Robust Anomaly Detection and Localization for Interleaved Unstructured Logs. IEEE Trans. Dependable Secure Comput. 20(4), 2762–2780 (2023). https://doi.org/10.1109/TDSC.2022.3162857

7. Xu, W., Huang, L., Fox, A., Patterson, D., Jordan, M.I.: Mining Console Logs for Large-Scale System Problem Detection. In: Proc. 3rd Workshop on Tackling Computer Systems Problems with Machine Learning Techniques (SysML), (2008).

8. Bai, S., Kolter, J.Z., Koltun, V.: An Empirical Evaluation of Generic Convolutional and Recurrent Networks for Sequence Modeling. arXiv preprint arXiv:1803.01271 (2018).

9. Vaswani, A., Shazeer, N., Parmar, N., et al.: Attention Is All You Need. In: Adv. Neural Inf. Process. Syst. (NeurIPS), vol. 30, pp. 5998–6008 (2017).

10. Le, V.H., Zhang, H.: Log-based Anomaly Detection with Deep Learning: How Far Are We? In: Proc. 44th Int. Conf. Softw. Eng. (ICSE), pp. 1356–1367 (2022). https://doi.org/10.1145/3510003.3510155

11. Albert, R.G.: System Logs Anomaly Detection. Are we on the right path? Appl. Artif. Intell. 39(1) (2025). https://doi.org/10.1080/08839514.2024.2440692

12. Zhu, J., He, S., He, P., Liu, J., Lyu, M.R.: Loghub: A Large Collection of System Log Datasets for AI-driven Log Analytics. In: Proc. 34th IEEE Int. Symp. Softw. Reliab. Eng. (ISSRE), pp. 355–366 (2023).

13. Du, M., Li, F.: Spell: Streaming Parsing of System Event Logs. In: Proc. IEEE Int. Conf. Data Min. (ICDM), pp. 859–864 (2016).

14. Liu, M., Wang, T., Zhao, Y., Xia, C., Zeng, Y., Tao, Y.: LogMUSE: Log Anomaly Detection via Multi-Scale Semantic Representation. IEEE Trans. Serv. Comput. 19(4), 2903–2916 (2026). https://doi.org/10.1109/TSC.2026.3700403

15. Dietz, S., Klede, K., Nguyen, A., Eskofier, B.M.: ContraLog: Log File Anomaly Detection with Contrastive Learning and Masked Language Modeling. arXiv preprint arXiv:2602.03678 (2026).

16. Liao, N., Liu, Z.: Log Anomaly Detection Method Based on Transformer and Temporal Convolutional Networks. IEEE Access 13, 68547–68560 (2025). https://doi.org/10.1109/ACCESS.2025.3561669